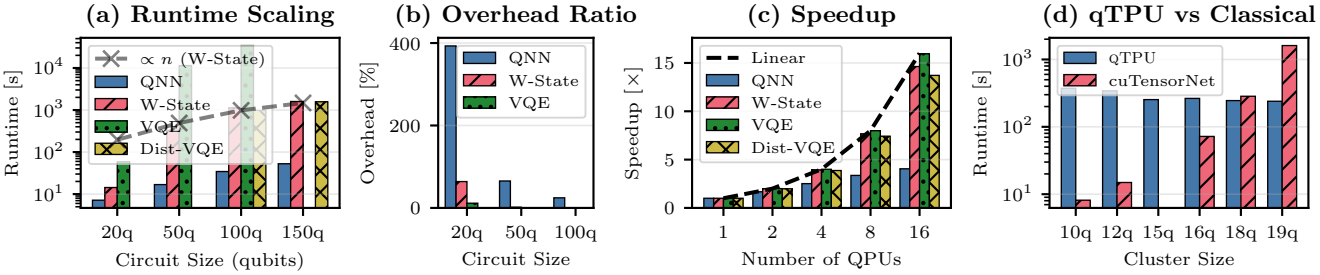
(a) Runtime Scaling
∝ n (W-State)
QNN
W-State
VQE
Dist-VQE
Runtime [s]
$10^4$
$10^3$
$10^2$
$10^1$
20q
50q
100q
150q
Circuit Size (qubits)
(b) Overhead Ratio
QNN
W-State
VQE
Overhead [%]
400
200
0
20q
50q
100q
Circuit Size
(c) Speedup
Linear
QNN
W-State
VQE
Dist-VQE
Speedup [×]
15
10
5
0
1
2
4
8
16
Number of QPUs
(d) qTPU vs Classical
qTPU
cuTensorNet
Runtime [s]
$10^3$
$10^2$
$10^1$
10q
12q
15q
16q
18q
19q
Cluster Size